

FixCity

When a streetlight breaks at midnight or a pothole blows out your tyre, who do you call? Nobody — because there's no system. We built one.

[LIVE PRODUCT](#)[NEXT.JS + SUPABASE](#)[AI-POWERED](#)[OPEN SUBMISSION](#)

60%

Civic complaints never get a reply.

People give up. They stop reporting. Broken things stay broken. That's not a data problem — it's a system problem. There is no system.

No structured reporting channel

PhoneCallsWhatsAppMessagesRandomTweets — all of it evaporates. Nothing is tracked, nothing is accountable.

Zero visibility after submission

Citizens submit a complaint and enter a black hole. Is it being looked at? Assigned? Done? Nobody knows — not even the officer.

Duplicate reports waste field teams

One broken manhole gets reported 20 times. A crew gets dispatched 20 times. Already-tight budgets tank even faster.

Password-only logins get shared

Officers share credentials. Technicians fake status updates. Without identity verification, the whole workflow is a fiction.

One city. One platform. Everyone in the loop.

FixCity is a live, deployed civic management system — not a prototype. Real users. Real reports. Real resolutions.



Citizen Portal

Report potholes, broken lights, leaks — with a photo and GPS pin. Track every status change in real time, and confirm when your issue is actually fixed.



Officer Dashboard

Ward and Taluk officers review complaints, assign technicians, and add resolution notes — with scoped, role-based access so nothing leaks across departments.



Technician Dashboard

A focused job queue. Go to site, fix it, mark resolved. Status updates propagate instantly — no chasing down officers to close a ticket.



AI Face Login

Officers and technicians authenticate via biometric face scan. No passwords to share. No fake updates. InsightFace runs on Hugging Face Spaces.



Smart Deduplication

AI groups duplicate reports by proximity and category. Officers see one consolidated job — not 15 copies of "pothole on MG Road."



Super Admin Console

City-wide analytics, officer management, and full oversight — all in one privileged dashboard with no friction.

From broken to fixed *in five steps.*

Every report follows an end-to-end pipeline. Every transition is timestamped. Nothing falls through the cracks.

STEP 01



Citizen Reports

Geo-tagged photo report — category, description, auto-detected location. Submitted in under 60 seconds from any device.

STEP 02



AI Checks for Duplicates

Report is screened against nearby open issues. If it's a duplicate, it's grouped — not sent as a separate ticket to clog the queue.

STEP 03



Officer Assigns

Officer reviews the priority queue, assigns the right technician. Face-verified login means it's actually the officer — not someone using their credentials.

STEP 04



Technician Resolves

Technician gets the job, goes to the site, marks it resolved. Status updates fire across all portals instantly — no manual follow-up.

STEP 05



Citizen Confirms

Citizen reviews the resolution. Accept it — report closes. Reject it with a reason — it re-enters the pipeline automatically.

Supabase realtime sync

Full audit trail

No manual follow-up needed

Built for how cities actually work.

Not a demo. Not a mockup. Every feature here is live and tested — including the AI bits.

5

distinct user roles, each with isolated access

6+

issue categories tracked with priority scoring

AUTH

Face Verification Login

InsightFace Buffalo_L model. 512-dim embeddings. Cosine similarity match. Staff can't share creds if there's nothing to share.

REPORTS

Geo-Tagged + Photo Reports

GPS auto-detect or manual input. Images go to Supabase Storage CDN. Works on mobile and desktop, no app install required.

INTELLIGENCE

Duplicate Detection Engine

Nearby reports with matching category get grouped under a parent. Officers see one job. Field teams make one trip.

LIFECYCLE

Resolution Review Loop

Citizens explicitly accept or reject resolved issues. Rejection auto-reopens the ticket and notifies the officer. Accountability that closes.

ACCESS

Multi-Tier Role Hierarchy

Super Admin → Taluk Officer → Ward Officer → Technician. Row Level Security in Postgres ensures nobody sees what they shouldn't.

ANALYTICS

Real-Time Analytics Console

Resolution rates, category breakdown, technician performance — charts that update live without a single manual refresh.

Tech that runs in production, not just on localhost.

Vercel edge network + Supabase Postgres + Hugging Face Docker space. All three running live.

FRONTEND

- Next.js 14 App Router
- TypeScript
- Tailwind CSS
- Framer Motion
- Shadcn UI + Lucide

BACKEND & DATA

- Supabase (Postgres)
- Supabase Storage CDN
- Row Level Security (RLS)
- Next.js API Routes
- Vercel Edge Deploy

AI / ML

- InsightFace (Buffalo_L)
- FastAPI (Python)
- Hugging Face Spaces
- Docker Container

FACE AUTH PIPELINE

Photo → InsightFace encoder → 512-dim embedding → cosine similarity → accept/reject in <1 second. No stored passwords for any staff account.

AI isn't a feature here. It's the backbone.

BIOMETRIC AUTH

Face Verification Login

No passwords to leak, share, or brute-force. Officers and technicians authenticate with their face — powered by state-of-the-art InsightFace embeddings deployed as a FastAPI microservice.

PRIORITY INTELLIGENCE

Dynamic Priority Scoring

Reports score based on category urgency, time pending, and duplicate count. Officers always see the most critical items first — the queue works, not just sorts alphabetically.

We used AI tools in development too — GitHub Copilot, Claude, ChatGPT — to move faster. Disclosed in full, as per FixForward's AI policy. AI helped us ship better, not differently.

SMART DEDUPLICATION

AI Duplicate Report Detection

Submitted reports are screened by proximity and category. Duplicates collapse under a parent record — reducing administrative noise and preventing repeat field dispatches to the same site.

AI-ASSISTED BUILD

Transparently AI-Assisted

Copilot + Claude accelerated our development. We're disclosing it upfront — it's a tool, not a shortcut. The architecture decisions, problem framing, and product choices are entirely ours.

Problems we solve.

Measurably.

5

user roles with isolated access & dashboards

100%

end-to-end traceable report lifecycle

<1s

face auth response time — production latency

Live

deployed on Vercel, right now, real data

WHO BENEFITS

 **Citizens** — report, track, confirm, reject

 **Ward Officers** — assign, monitor, update

 **Taluk Officers** — departmental oversight

 **Technicians** — focused task queue

 **Super Admins** — city-wide analytics

PROBLEM → SOLUTION

→ Untracked complaints → Full audit trail

→ No accountability → Role-based audit log

→ Fake officer logins → Biometric verification

→ Duplicate work orders → AI deduplication

→ Unresolved complaints → Citizen review loop

Where we are. Where we're going.

Phase 1 is shipped and live. Phase 2 is in progress. Phase 3 is the scale play.

PHASE 1 — SHIPPED ✓

Core Platform

- Citizen geo-tagged reporting with photo upload
- Officer & technician dashboards
- Full status lifecycle — Pending to Closed
- AI face verification login (InsightFace)
- Supabase DB, Storage, and RLS security
- Vercel production deployment — live now
- Citizen resolution review and rejection flow

PHASE 2 — IN PROGRESS ◆

Intelligence Layer

- AI duplicate detection engine
- Priority scoring algorithm per report
- Real-time analytics dashboard
- Super Admin city-wide console
- Multi-tier role hierarchy (Taluk → Ward)
- Ward officer management controls

PHASE 3 — PLANNED ◇

Scale & Outreach

- WhatsApp / SMS status notifications
- Mobile-first PWA app
- AI chatbot for citizen self-service
- Predictive maintenance alerts
- Multi-city / multi-municipality support
- Open API for 3rd-party integrations

Fix the city. One report at a time.

FixCity makes urban issue resolution transparent, accountable, and closed-loop — with AI where it matters and humans where it counts.

 fixcity-chi.vercel.app

 github.com/RonnyHamilton/Fixcity

✓ AI tools used: GitHub Copilot · Claude · ChatGPT
Disclosed per FixForward AI Usage Policy

BUILT WITH

Next.js 14

App Router + TypeScript

Supabase

Postgres + Storage + RLS

InsightFace

AI biometric auth

Vercel

Edge deployment, live now



fixcity-chi.vercel.app
scan or type to try it live